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- CLASS : SYIT
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6a. Write a Python program to import a CSV dataset using Pandas and display the first 5 and last 5 records.

```
import pandas as pd
```

pd.read_csv() – reads a CSV dataset.

```
df = pd.read_csv("Students.csv")
```

Print top 5 values

```
print(df.head())
```

	Roll No	Name	Department	Marks	Attendance
0	101	Amit	BSc IT	78	85
1	102	Priya	BSc CS	65	90
2	103	Rahul	BSc IT	88	92
3	104	Neha	BSc CS	72	80
4	105	Rohan	BSc IT	55	70

Print last 5 values

```
print(df.tail())
```

	Roll No	Name	Department	Marks	Attendance
0	101	Amit	BSc IT	78	85
1	102	Priya	BSc CS	65	90
2	103	Rahul	BSc IT	88	92
3	104	Neha	BSc CS	72	80
4	105	Rohan	BSc IT	55	70

6b. Write a Python program to analyze the dataset by displaying the number of rows and columns, column names, data types, and basic statistical information.

```
print(df.shape) #number of rows and columns
```

```
(5, 5)
```

```
print(df.columns)
```

```
Index(['Roll No', 'Name', 'Department', 'Marks', 'Attendance'], dtype='object')
```

```
print(df.dtypes)
```

```
Roll No      int64
Name         object
Department   object
Marks        int64
Attendance   int64
dtype: object
```

```
print(df.describe)
```

```
<bound method NDFrame.describe of
0      101  Amit  BSc IT  78      85
1      102  Priya BSc CS  65      90
2      103  Rahul BSc IT  88      92
3      104  Neha  BSc CS  72      80
4      105  Rohan BSc IT  55      70>
```

6c. Write a Python program to filter records from the dataset using conditions. For example, display students having marks greater than 60 or employees having salary greater than 40,000.

```
result = df[df["Attendance"]>=90]
print(result)
```

```
Roll No  Name Department  Marks  Attendance
1      102  Priya      BSc CS    65          90
2      103  Rahul      BSc IT    88          92
```

6d. Write a Python program to sort the dataset based on a selected column and display the top 5 records. For example, sort students by marks or products by price.

```
result = df.sort_values("Marks")
print(result)
```

```
Roll No  Name Department  Marks  Attendance
4      105  Rohan      BSc IT    55          70
1      102  Priya      BSc CS    65          90
3      104  Neha      BSc CS    72          80
0      101  Amit      BSc IT    78          85
2      103  Rahul      BSc IT    88          92
```

6e. Write a Python program to perform data analysis on the dataset by finding the maximum, minimum, average, and total values of a numerical column.

```
print("Maximum Marks ", df["Marks"].max())
```

```
print("Maximum Marks ",df["Marks"].max())  
print("Minimum Marks ",df["Marks"].min())  
print("Avearge Marks ",df["Marks"].mean())  
print("Total Marks ",df["Marks"].sum())
```

```
Maximum Marks    88  
Minimum Marks    55  
Avearge Marks    71.6  
Total Marks     358
```